

User Guide

Zero-Inflated Negative Binomial (ZINB) Latent-Class Trajectories

File: zinb.sas Traj2 (prototype, under QA) Author: Anum Zafar

Prototype. This model is undergoing quality assurance. Interfaces and results may change before release.

What it does

This file fits zero-inflated negative binomial (ZINB) latent-class trajectory models for count outcomes in PROC NLMIXED. It sorts subjects into a few latent classes, each with its own count trajectory over time. ZINB extends the zero-inflated Poisson model with a class-specific dispersion parameter, so it can absorb overdispersion (variance well above the mean), which is common in high-utilization Medicare count data. Everything is macro-only with no compiled code, so it runs as-is in the CMS VRDC and any SAS 9.4 or newer session.

What you need

- SAS 9.4 or newer with SAS/STAT (PROC NLMIXED).
- ODS Graphics for the plots (%ct_zinb_plots). The fit itself runs without them.
- A wide, one-row-per-subject table of counts, and write access to WORK.

The model in brief

Each class c has a mean count trajectory on the log scale, a zero-inflation probability, and its own dispersion. The dispersion k is estimated on the log scale ($k = \exp(\log k)$) so it stays positive.

```
mean (log link):  mu_c(t) = exp(beta0_c + beta1_c*t + ...)
zero-inflation:   p_c(t)  = logistic(gamma0_c + gamma1_c*t + ...)
dispersion:       k_c = exp(logk_c),    Var = mu + mu^2/k

P(Y=0 | c) = p + (1-p) * (k/(k+mu))^k
P(Y=y | c) = (1-p) * NB2(y; mu, k),    y > 0

class share:      w_c = exp(alpha0_c) / (1 + sum_k exp(alpha0_k))
```

Class A is the mixing reference (it has no α_0). As k grows, ZINB approaches ZIP. The fit uses a numerically stable log-sum-exp both for the zero component and for mixing across classes. Missing Y values are skipped in the likelihood, valid under missing-at-random.

Quick start (simulated data)

The simulator builds BASE_FILE_SRS for you, then the fit and plot run. Set starting values first if you want them (see below); anything you leave unset starts at 0.

```
%let T = 12;

/* 1. Simulate 3-class ZINB counts and build BASE_FILE_SRS */
%sim_data(class=3, n=500, T=&T, seed=1, miss_pattern=balanced);

/* 2. Fit a 3-class, quadratic-mean, constant-inflation model */
%ct_zinb_nlmixed(
  data=BASE_FILE_SRS, id=BENE_ID,
```

```
yvars=SUM_Q1-SUM_Q12,
nclass=3, class_labels=A B C,
order=2, p_order=0, T=&T);

/* 3. Plot per-class curves + mixture mean */
%ct_zinb_plots(pe=pe_zinb, T=&T);
```

The settings you edit

Argument	Meaning
data / id	Input wide table and subject id (defaults BASE_FILE_SRS, BENE_ID).
yvars	Count outcome columns, one per time point, in time order (e.g. SUM_Q1-SUM_Q12).
nclass	Number of latent classes.
class_labels	Space-separated labels; the count must equal nclass or the macro stops.
order	Mean-curve degree: 1 linear, 2 quadratic, 3 cubic.
p_order	Zero-inflation degree: 0 constant, 1 linear, and so on.
T	Number of time points.
tech / maxiter	Optimizer technique (default newrap) and iteration cap (default 500).
pe_out / fit_out	Output dataset names for estimates and fit statistics.
bounds	Optional parameter bounds for stability.

Setting starting values

Unlike the other Traj2 models, starting values here are set with %let lines in section [C] of the file, before you call the fit. Any parameter you do not set starts at 0. Names follow the pattern below; class A has no alpha0 because it is the reference. A logk near 0 (which means $k = 1$) is a reasonable warm start.

```
%let alpha0_B = -0.50; /* mixing logits, class B onward */
%let beta0_A = -0.40; /* mean-curve coefficients: beta0/1/2/3_{class} */
%let beta1_A = 0.06;
%let gamma0_A = -1.20; /* zero-inflation logits: gamma0/1/2/3_{class} */
%let logk_A = 0.00; /* dispersion: k = exp(logk) */
```

Using your own data

Point data= at a wide table with one row per subject. The fit does not read a separate time-index column: time is taken from the column order of yvars, so list the count columns in time order.

Column	What to supply
Subject id	Named by id, unique per subject.
Counts	Named by yvars, one non-negative integer column per time point, in time order.

- Counts must be non-negative integers. Excess zeros are expected and are what the zero-inflation part handles.
- Match class_labels to nclass. A mismatch stops the macro right away by design.

Running and choosing the number of classes

Each call fits one model for a fixed nclass. To choose the number of classes, refit with different nclass values (and matching class_labels) and compare BIC in the fit-statistics dataset, preferring the lowest BIC with sensible class sizes.

What you get back

- `pe_zinb`: parameter estimates and standard errors (alpha, beta, gamma, logk).
- `fit_zinb`: fit statistics (AIC, BIC, -2 log L) for comparing candidate class counts.
- `traj_zinb` and `mix_zinb`: the per-class ZINB mean curves and the mixture-mean series.
- Two plots from `%ct_zinb_plots`: per-class trajectories, and the same with the mixture mean overlaid.

Troubleshooting

Symptom	Fix
Macro stops immediately	nclass and class_labels counts differ. Make them equal.
Will not converge	NB fits are sensitive. Start logk near 0, set starting values in section [C], add bounds, or try fewer classes or a lower order.
Zero-inflation is flat over time	p_order=0 holds it constant. Raise p_order for a time trend in the structural zeros.
Dispersion k very large	Little overdispersion; the model is near ZIP, and ZIP may be enough.

CHAO Lab, Rutgers. Full parameter names and macro details are in the data dictionary for zinb.sas.